PLC OUTPUT INTERLOCK MATRIX

PLC Output	Device / Function	Permitted States	ON Condition	Forced OFF Condition	Reference
P20	MTR-01 Conveyor Run	S30, S90	Safety Ready; transfer condition active	P00 OFF, P07 OFF, P02 Stop, SF0, PE-02/PE-03 reached	DWG 008 / SHT 1
P21	MTR-01 Direction	S30, S90	OFF = Forward Default	P20 OFF or SF0	DWG 008 / SHT 2
P22	SOL-01 Test Clamp	S40, S50, S60	Clamp close sequence active	P00 OFF, P07 OFF, P02 Stop, SF0, S70 Release	DWG 008
P23	SOL-02 PASS Gate	S81-P, S90	D0020 = 1; P24 = OFF AND P25 = OFF P09 = ON CONFIRMS PASS GATE POSITION	P00 OFF, P07 OFF, P02 Stop, SF0, S100 Gate Home	DWG 008
P24	SOL-03 FAIL Gate	S81-F, S90	D0020 = 2; P23 = OFF AND P25 = OFF P0A = ON CONFIRMS FAIL GATE POSITION	P00 OFF, P07 OFF, P02 Stop, SF0, S100 Gate Home	DWG 008
P25	SOL-04 RETEST Gate	S81-R, S90	D0020 = 3; P23 = OFF AND P24 = OFF P0B = ON CONFIRMS RETEST GATE POSITION	P00 OFF, P07 OFF, P02 Stop, SF0, S100 Gate Home	DWG 008
P26	LT-01-G Green Lamp	S10~S110	Safety Ready; No Fault; Auto Ready/Run	S00 or SF0	DWG 008
P27	LT-01-Y Yellow Lamp	Manual / Wait / Warning	Manual Mode or Controlled Stop or Warning	S00, S10~S90 Normal Auto, or SF0	DWG 008
P28	LT-01-R Red Lamp	S00, SF0	Safety Not Ready or Process Fault	Safety Ready and Fault Cleared	DWG 008
P29	BZ-01 Buzzer	SF0	Fault / Alarm Active	P03 Reset after Fault Cause Cleared	DWG 008

NOTE:
P23, P24, AND P25 SHALL BE MUTUALLY EXCLUSIVE.
ONLY ONE SORT GATE OUTPUT MAY BE ON AT ANY TIME.

NOTE:
P20 SHALL NOT BE ON WHILE P22 CLAMP CLOSE COMMAND
IS ACTIVE, EXCEPT WHERE SPECIFICALLY DEFINED BY SEQUENCE LOGIC.

MCU-01 TEST HANDSHAKE / REGISTER MATRIX

Register	Signal	Direction	Value	PLC / MCU Action	Valid State
D0020	TEST_RESULT_CODE	MCU → PLC	0	No Result / Test In Progress	S60
D0020	TEST_RESULT_CODE	MCU → PLC	1	PASS Result	S60 → S80
D0020	TEST_RESULT_CODE	MCU → PLC	2	FAIL Result	S60 → S80
D0020	TEST_RESULT_CODE	MCU → PLC	3	RETEST Result	S60 → S80
D0020	TEST_RESULT_CODE	MCU → PLC	99	Invalid Result / Test Fault → SF0	S60
D0021	TEST_COMPLETE_FLAG	MCU → PLC	0	Test Incomplete / Result Not Valid	Before Complete
D0021	TEST_COMPLETE_FLAG	MCU → PLC	1	Result Valid / Test Complete	S60 → S70
D0022	TEST_REQUEST_FLAG	PLC → MCU	0	Test Request Inactive / Clear	S10~S50, S70~SF0
D0022	TEST_REQUEST_FLAG	PLC → MCU	1	Start Test / Evaluate Test Condition	S60
D0023	MCU_HEARTBEAT_COUNTER	MCU → PLC	0~65535	MCU increments every 500 ms	Any Auto State
T0007	MCU_COMMS_WATCHDOG	PLC Internal	3 s	D0023 unchanged → ALM-108 / SF0	S30~S90

TEST HANDSHAKE SEQUENCE:

1. PLC ENTERS S60 TEST EXECUTION.
2. PLC WRITES D0022 = 1 TEST REQUEST.
3. MCU-01 PERFORMS TEST EVALUATION.
4. MCU-01 WRITES D0020 = 1, 2, 3, OR 99.
5. MCU-01 WRITES D0021 = 1 ONLY AFTER D0020 IS VALID.
6. PLC VALIDATES D0021 = 1 AND D0020 = 1 / 2 / 3 / 99.
7. PLC WRITES D0022 = 0 ON EXIT FROM S60.
8. MCU-01 CLEARS D0021 = 0 AND D0020 = 0 BEFORE NEXT TEST.

PLC SHALL EVALUATE D0020 TEST RESULT CODE
ONLY WHEN D0021 TEST COMPLETE FLAG = 1.

[GENERAL CONTROL NOTES]

[1] THIS DRAWING DEFINES PLC-01 OUTPUT INTERLOCKS AND MCU-01 TEST HANDSHAKE LOGIC.

[2] HARDWIRED SAFETY FUNCTION IS SHOWN ON DWG 006. PLC STATE LOGIC SHALL NOT BE THE PRIMARY SAFETY FUNCTION.

[3] P00 IS SR-01 SAFETY HEALTHY MONITOR. P07 IS DR-01-AUX GUARD DOOR CLOSED MONITOR.

[4] P23, P24, AND P25 SHALL BE MUTUALLY EXCLUSIVE. ONLY ONE SORT GATE OUTPUT MAY BE ON AT ANY TIME.

[5] PLC SHALL EVALUATE D0020 TEST RESULT CODE ONLY WHEN D0021 TEST COMPLETE FLAG = 1.

[6] PLC SHALL WRITE D0022 = 1 ONLY DURING S60 TEST EXECUTION.
D0022 SHALL BE RESET TO 0 ON EXIT FROM S60 OR SF0 ENTRY.

[7] ALARM, TIMEOUT, INDICATOR, AND BUZZER LOGIC: SEE DWG 012.

ALARM / FAULT MATRIX, TIMER SETTING MATRIX,
AND INDICATOR / BUZZER STATUS MATRIX:
SEE DWG 012.

NO.	DATE	REVISION
A	2026-09-10	INITIAL RELEASE

DWG TITLE
EDS Wafer Map Linked Package
Final Test and Multi-Bin Sorting Cell

PLC OUTPUT INTERLOCK
AND MCU TEST HANDSHAKE

DRAWN BY: J. H. KO
DATE: 2026-09-10
DWG NO: 011
SHEET NO: 2 OF 2